Malaria — Diagnosis, Treatment and Prevention

1. Epidemiology and Global Burden

Malaria remains a major global public health problem, with the WHO World Malaria Report 2022 reporting 247 million cases and 619,000 deaths globally in 2021. Sub-Saharan Africa bears 95% of the global burden, but South Asia — particularly India — contributes significantly.

India accounts for approximately 83% of malaria cases in the WHO South-East Asia Region. Odisha, Jharkhand, Chhattisgarh, Madhya Pradesh, and North-Eastern states have the highest transmission. India contributes substantially to Plasmodium vivax burden — the globally neglected second-most-common malaria species.

India achieved WHO certification of malaria elimination in several states (e.g., Sri Lanka in the region eliminated malaria in 2016). India's National Strategic Plan for Malaria Elimination aims to eliminate malaria from 21 states and 4 union territories by 2022, and from the entire country by 2030.

2. Parasitology and Life Cycle

Five Plasmodium species infect humans: P. falciparum (most dangerous, predominates in Africa and coastal India), P. vivax (most widespread, predominates in India and Asia — produces dormant liver stage), P. malariae (causes quartan malaria, persists decades), P. ovale (rare, West Africa), P. knowlesi (zoonotic, Southeast Asia — can cause severe malaria).

Life cycle: female Anopheles mosquito injects sporozoites during blood meal → sporozoites migrate to liver (hepatic stage, no symptoms) → multiply into thousands of merozoites → merozoites released into bloodstream → infect red blood cells (RBCs) → asexual multiplication (ring → trophozoite → schizont → merozoite rupture) → RBC rupture causes fever cycle.

P. vivax and P. ovale produce hypnozoites — dormant liver forms that can reactivate months to years after the primary infection, causing relapsing malaria. Only primaquine/tafenoquine can eliminate hypnozoites (radical cure).

The sexual stage (gametocytes) circulates in the blood and is taken up by mosquitoes during feeding, completing the transmission cycle. P. falciparum gametocytes persist for weeks even after treatment, maintaining transmission potential.

3. Clinical Features

Classic malaria paroxysm: cold stage (rigors, chills, severe shaking, 1-2 hours) → hot stage (high fever 39-41°C, headache, vomiting, 2-6 hours) → sweating stage (drenching sweats, fever breaks, exhaustion). Cycle repeats every 48 hours (P. vivax, P. falciparum) or 72 hours (P. malariae).

Other symptoms: headache, myalgia, arthralgia, anorexia, nausea, vomiting, abdominal pain, splenomegaly (enlarged spleen — chronic malaria), anaemia (pallor, fatigue, dyspnea) due to destruction of infected and uninfected RBCs.

Uncomplicated malaria: above symptoms without evidence of end-organ dysfunction. Severe malaria (WHO criteria): cerebral malaria (impaired consciousness, seizures — most important criterion), severe anaemia (Hb <7 g/dL in adults), acute kidney injury, pulmonary oedema/ARDS, spontaneous bleeding, hypoglycemia, metabolic acidosis, haemoglobinuria (blackwater fever), hyperparasitaemia (>5% infected RBCs).

Cerebral malaria: diffuse encephalopathy — symmetrical, no focal signs. Coma (Blantyre/Glasgow Coma Scale). Seizures (especially children). Retinal haemorrhages and papilloedema on fundoscopy. Poor prognostic sign: abnormal posturing (decerebrate/decorticate), absence of corneal reflex.

4. Diagnosis

ntiation	Sensitivity	Availability
	High when done well	All levels
oLDH (pan)	Moderate-High	PHC level
ard)	Highest	Reference labs
	High	Increasing
ly	High	Field use

Blood smear remains the gold standard for field use and species identification. Examine thick smear (parasite density) and thin smear (morphology) of Giemsa-stained blood. In Odisha-type high-transmission settings, RDTs enable frontline health workers to diagnose and treat at village level.

Important: P. falciparum pfhrp2/3 gene deletions can cause false-negative HRP2 RDTs. Significant in India's North-Eastern states. Use pLDH-based RDTs or smear when clinical suspicion is high despite negative HRP2 RDT.

5. Treatment

P. falciparum Treatment

Artemisinin-based Combination Therapy (ACT) is the WHO-recommended treatment for all uncomplicated P. falciparum malaria. Options: Artemether-lumefantrine (AL, Coartem — 6 doses over 3 days), Artesunate-amodiaquine (ASAQ), Artesunate-mefloquine (ASMQ — used in South-East Asia, drug resistance concerns), Dihydroartemisinin-piperaquine (DHA-PPQ).

India's NVBDCP recommends artesunate-sulfadoxine-pyrimethamine (AS+SP) in most states. Artemisinin resistance is emerging in South-East Asia (Cambodia, Myanmar) and must be monitored.

Severe P. falciparum malaria: IV artesunate 2.4 mg/kg at 0, 12, 24 hours, then every 24 hours. Superior to quinine. Start within 1 hour of diagnosis. Switch to oral ACT once patient can take oral medication. Blood transfusion for severe anaemia.

P. vivax Treatment and Radical Cure

Chloroquine (10 mg base/kg day 1, 10 mg base/kg day 2, 5 mg base/kg day 3) remains effective for most P. vivax in India (some resistance reported in East India and Indonesia).

Radical cure with primaquine: 0.25 mg base/kg/day for 14 days (standard dose) eliminates liver-stage hypnozoites and prevents relapse. G6PD testing mandatory before primaquine to prevent haemolytic anaemia (G6PD deficiency common in tribal populations in India).

Tafenoquine (single dose, 300 mg): new radical cure option approved in India. More convenient but higher haemolytic risk in G6PD-deficient individuals. Quantitative G6PD testing required.

6. Severe Malaria Emergency Protocol

Severe malaria is a medical emergency with mortality of 15-25% even with treatment. Immediate hospitalization required.

Establish IV access. Start IV artesunate immediately (do not wait for test results if clinical picture suggests severe malaria). Monitor blood glucose (malaria causes hypoglycemia; artesunate also lowers glucose). Give dextrose if hypoglycemic.

Seizure management: IV diazepam 0.15 mg/kg (children) or lorazepam. Maintain airway. Prophylactic anticonvulsants NOT recommended. In cerebral malaria with seizures: phenobarbital IM can prevent recurrence.

Fluid management: careful — there is narrow margin between underhydration (hypotension, AKI risk) and overhydration (pulmonary oedema in severe malaria). Transfuse packed RBCs for Hb <7 g/dL or haematocrit <20%.

Dexamethasone is NOT recommended in cerebral malaria (increases coma duration and GI bleeding). Exchange transfusion is no longer recommended.

7. Prevention — Individual and Community

Vector control: insecticide-treated bed nets (ITNs), especially long-lasting insecticidal nets (LLINs), are the most cost-effective individual protection. India distributes LLINs through NVBDCP in high-burden districts. Indoor Residual Spraying (IRS) of houses with insecticides is a key programme intervention.

Personal protection: use DEET or picaridin-based repellents (reapply every 4-6 hours). Wear full-sleeve, light-coloured clothing at dusk and dawn when Anopheles mosquitoes are most active. Protective screens on windows and doors.

Chemoprophylaxis for travellers to endemic areas: doxycycline (100 mg/day, start 1 day before travel, continue 28 days after), atovaquone-proguanil/Malarone (start 1-2 days before, continue 7 days after), or mefloquine (weekly, start 2-3 weeks before — not recommended in all areas due to resistance/side effects).

No approved malaria vaccine available in India yet. RTS,S/AS01 (Mosquirix) vaccine recommended by WHO in 2021 for African children; approved in several African countries. Phase 3 trial of R21/Matrix-M shows 75-80% efficacy. These vaccines target the pre-erythrocytic stage of P. falciparum.

8. Malaria in Pregnancy

Malaria in pregnancy is particularly dangerous: P. falciparum sequestration in the placenta causes placental malaria, leading to intrauterine growth restriction (IUGR), low birth weight, maternal anaemia, and increased risk of maternal and neonatal death.

Treatment: Artemether-lumefantrine is safe in 2nd and 3rd trimesters. In 1st trimester (when organogenesis is most vulnerable), quinine + clindamycin is recommended despite its toxicity (AL may be used if quinine not available — GRADE evidence evolving). Chloroquine for P. vivax is safe in all trimesters.

Primaquine and tafenoquine are CONTRAINDICATED in pregnancy (haemolytic risk to fetus with unknown G6PD status). P. vivax patients may only receive chloroquine during pregnancy; radical cure is deferred until after delivery and end of breastfeeding.

IPTp (Intermittent Preventive Treatment in Pregnancy): sulfadoxine-pyrimethamine (SP) given at antenatal visits for P. falciparum prevention in high-transmission Africa. Not routinely implemented in India.

9. Complications and Sequelae

Post-malaria neurological syndrome: rare syndrome following cerebral malaria — neuropsychiatric symptoms, ataxia, psychosis. Usually self-limiting over weeks.

Tropical splenomegaly syndrome (hyperreactive malarial splenomegaly): massive splenomegaly due to aberrant immunological response to repeated infections. Risk of rupture. Long-term antimalarial prophylaxis required.

Nephrotic syndrome: associated with P. malariae — rare, can cause membranoproliferative GN. Chloroquine does not prevent this; management is primarily supportive.

Long-term outcomes after severe malaria in children: neurological sequelae (cognitive impairment, epilepsy, behaviour problems) occur in 25% of children surviving cerebral malaria. Rehabilitation and educational support are essential.

10. Challenges in India and the Road to Elimination

Major challenges: tribal and forest-dwelling populations in Odisha and Jharkhand with limited healthcare access; mobile and migrant worker populations (construction, mining) with high exposure and low healthcare access; asymptomatic malaria reservoirs; emerging artemisinin resistance watch.

Surveillance: India's HMIS-malaria reporting system. API (Annual Parasite Incidence) and SPR (Slide Positivity Rate) are key indicators. Recent focus on case-based surveillance — Nikshay for malaria is being expanded.

India's elimination strategy: universal access to early diagnosis and prompt treatment (EDPT) within 24 hours of fever onset; scale up RDT use at village level; aggressive IRS and LLIN distribution; inter-sectoral coordination (tribal welfare, forest department).

Indoor residual spraying: selective IRS in high-burden areas has been highly effective. Resistance to DDT (organochlorine) prompted shift to malathion and newer pyrethroids. Resistance monitoring is critical to maintaining IRS effectiveness.